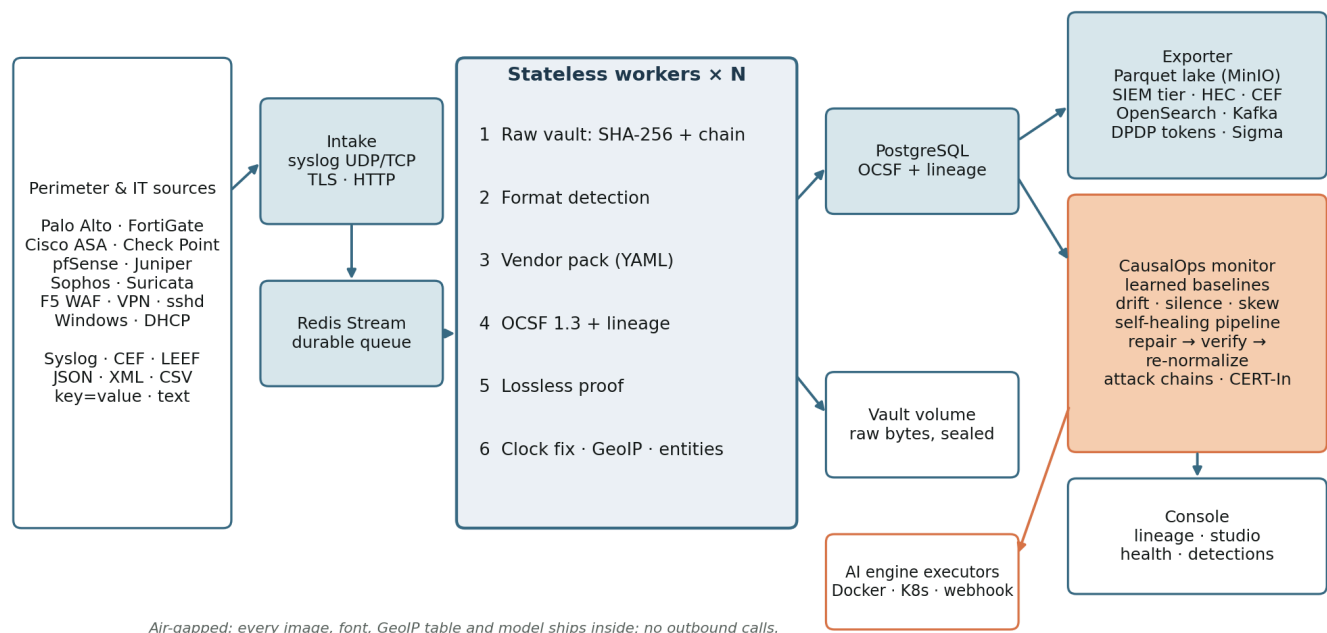


CausalOps ULPF: Universal Log Pre-processing Framework

SIH 2026 · PS 26156 (NTRO) · Blockchain & Cybersecurity · Architecture document



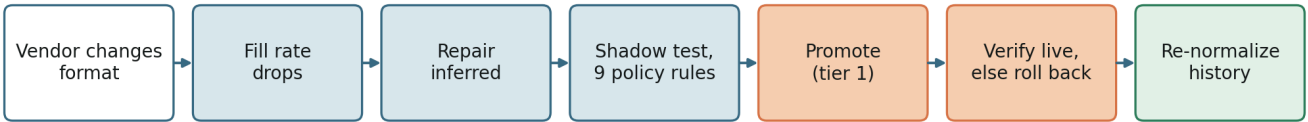
Data path (every event)

Intake accepts syslog over UDP, TCP (newline or RFC 6587 octet counting) and TLS, and HTTP. Messages are queued unchanged in a durable Redis stream. **Stateless workers** (a consumer group; add replicas to scale) first append the raw bytes to a **write-once vault** with a SHA-256 per event and a hash chain per segment, then detect the format, apply the source's **vendor pack** (declarative YAML: matching, field extraction, mapping, value translation), normalize to **OCSF 1.3** and keep every vendor field under `unmapped`. The **lossless proof** rebuilds the raw event from the normalized record alone and compares its SHA-256 with the vault. Clock correction, offline GeoIP and the cross-vendor entity graph are added before the record is committed with its lineage (uid, vault segment, offset, chain hash, pack version). A crashed worker's events are reclaimed by the others, so the conservation check *received = stored + in flight* holds.

How PS 26156 is met

Requirement	Implementation
a Preserve raw data	Write-once vault, SHA-256 + hash chain, retention-safe anchors; lossless proof per event
b Parse source attributes	14 packs (12 vendors + generic CEF/LEEF), exact field spans, Drain3 for unknown formats
c Common taxonomy	OCSF 1.3 classes 2004 · 3002 · 4001 · 4002 · 4003 · 4004, vendor fields kept in unmapped
d Traceability	Every record carries uid, raw SHA-256, vault position, pack@version; click-through lineage view
e Plug-and-play onboarding	Parser Studio: samples → proposed pack → test vs champion → activate (live in 10 s, no restart)
f Unified visibility	Console: sources, events, entity graph across vendors, pipeline health, detections
g SIEM / data lake	Parquet lake on MinIO (S3), SIEM tier, Splunk HEC, CEF syslog, OpenSearch, Kafka
h AI/ML ready	OCSF Parquet features; CausalOps anomaly gate, causal ordering, 10 Sigma rules on OCSF
i Less parser effort	YAML instead of code; automatic proposals; automatic repair after format changes
j Air-gapped	All images, fonts, GeoIP (DB-IP Lite) bundled; offline image archive; no outbound calls
k Containers	One docker compose; images on Docker Hub; offline tarball for isolated networks

What no log pipeline usually does: CausalOps watches the pipeline itself



Raw bytes never change; every repaired event is proven lossless again and keeps its previous revision.

Each source's parsing quality (share of expected OCSF attributes present), volume and clock offset are learned from its own quiet history with robust baselines (median, MAD) and a threshold calibrated to at most 1% false alarms. A vendor format change becomes a **parser-drift incident** naming the lost attributes and the new fields; the repair is inferred, shadow-tested against the current pack on drifted and earlier events, promoted through a tiered policy (kill switch, dry run, tier, fixes the drift, no regression, lossless, evidence, cooldown, reversible), verified on live data or rolled back, and the broken window is **re-normalized from the vault**. Silent sources, clock skew and crashed workers are handled the same way.

Twelve differentiators

1	Parser drift detection with verified, reversible pack promotion	7	Clock-skew estimation and correction per source
2	Provable losslessness (rebuild from OCSF = vault SHA-256)	8	SIEM cost tiering with lake pointers, no data lost
3	Silent-source detection from learned volume	9	DPDP pseudonymised export, audited detokenisation
4	Self-healing pipeline through the Docker executor	10	Sigma rules on OCSF: one rule, every vendor
5	Historical re-normalization with revision history	11	Cross-vendor entity graph and attack-chain correlation
6	CERT-In 2022 compliance from measurements	12	Ed25519-signed bundles for data diodes and evidence

Measured on the running system

Events stored / proven lossless	812,155 / 812,155 (100.0% normalized)
Processing throughput, one worker	2,684 events/s ≈ 232 M events/day per worker, lossless 100.0%
Parser drift (format change)	3 resolved; median detect 30 s, resolve 122 s
SIEM tier	59.1% fewer bytes than the full stream (5.1 → 2.1 GB/day at this rate)
CERT-In checks	5 passed, 1 failed (measured, not assumed)

Deployment

One command (`bash scripts/demo/start.sh`) brings up intake, workers, monitor, exporter, MinIO, Redis, PostgreSQL, the CausalOps platform and the console. For isolated networks `scripts/release/export_offline.sh` saves every image to one archive and `load_offline.sh` starts the stack without contacting any registry. Workers scale with `--scale ulpf-worker=N`; Kafka and OpenSearch run under `--profile full`.